Program : **Diploma in Robotics Process Automation**	
Course Code : **3307**	Course Title: **Problem Solving Using Python Lab**
Semester : **3**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To learn and understand the basic concept of Python programming and use it in real world problem solving.
- To learn the implementation of data structures like, Lists, Tuples and Dictionaries.
- To implement operation on files and exception handling.
- To implement a micro project using Python

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic functions and features of Computer, Operating system and Internet applications.		Introduction to IT systems Lab	1
Basic programming skills in C language.		Introduction to Computation and Programming	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Make use of basic operators and control structures in Python programming.	**10**	Applying
CO2	Develop Python programs using user-defined functions and strings.	**11**	Applying
CO3	Demonstrate the use of built-in data structures like Lists, Tuples and Dictionaries.	**9**	Applying
CO4	Implement the operations on files and exception handling methods.	**12**	Applying
	Series Test	**3**	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3	3			
CO2	3	3	3	3			
CO3	3	3	3	3			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	**Name of Experiment**	**Duration (Hours)**	**Cognitive Level**
CO1	**Make use of basic operators and control structures in Python programming.**		
M1.01	Demonstrate simple problems for performing input and output functions.	1	Applying
M1.02	Solve simple problems using operators	1	Applying
M1.03	Solve problems using decision making statements.	4	Applying
M1.04	Develop programs for solving iterative statements	4	Applying
CO2	**Develop Python programs using user-defined functions and strings.**		
M2.01	Implement programs to solve problems using functions	5	Applying
M2.02	Construct programs using recursive functions.	3	Applying
M1.03	Solve problems using string operations.	3	Applying
	Series Test 1	1.5	
CO3	**Demonstrate the use of built-in data structures like Lists, Tuples and Dictionaries.**		
M3.01	Write programs for performing List operations-Traversal	3	Applying
M3.02	Construct programs for performing Tuples operations.	3	Applying
M3.03	Develop programs for creating Dictionaries and perform Dictionar operations	3	Applying

CO4	Implement the operations on files and exception handling methods		
M4.01	Write programs to perform operations on files- opening, closing, reading, writing etc.(Text file, Binary file, CSV file).	7	Applying
M4.02	Develop programs for implementing pickle module.	2	Applying
M4.03	Develop programs for handling exceptions.	3	Applying
	Series Test 2	1.5	

**** - Micro Project**

Students are expected to do a micro project using Python during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record.

Example: Develop program such as

- Dice roll simulator,
- Guess the number game,
- Random password generator

Text / Reference:

T/R	Book Title/Author
T1	Balagurusamy E, Introduction to Computing and Problem Solving Using Python.
R1	Lambert K.A, Fundamentals of Python-First Programs. Cengage Learning India 2015.
R2	Perkovic .L., Introduction to Computing Using Python, 2nd Edition
R3	Zelle. J., Python Programming: An Introduction to Computer Science.
R4	Allen Downey, Jeffrey Elkner, Chris Meyers., How to think like a computer scientist: learning with Python, 1st Edition

Online Resources:

Sl.No	Website Link
1	https://www.tutorialspoint.com/python/index.htm
2	https://www.javatpoint.com/python-tutorial
3	https://www.programiz.com/python-programming
4	https://www.w3schools.com/python/default.asp

<u>List of Sample Experiments</u>

1. Write a python program to find the sum of 2 numbers.
2. Write a python program to find the area of a circle.
3. Write a python program to find the area and perimeter of a rectangle.
4. Write a python program to swap 2 numbers using temporary variable.
5. Write a python program to swap 2 numbers without using temporary variable.
6. Write a python program to convert celcius to farenheit.
7. Write a python program to calculate simple interest.
8. Write a python program to check whether the given number is divisible by 2 or 3.
9. Write a python program to find the largest of 2 numbers.
10. Write a python program to find the largest of 3 numbers.
11. Write a python program to find whether the given number is even or not.
12. Write a python program to find whether the given number is positive or negative.
13. Write a python program to find the largest of n numbers.
14. Write a python program to find the sum of n numbers.
15. 1Write a python program to find the factorial of a number.
16. Write a python program to display fibonacci series.
17. Write a python program to find the sum of a sequence of numbers between a lower bound and upper bound.
18. Write a python program to find factorial using function.
19. Write a python program to reverse a number using function.
20. Write a python program to find the sum of digits of a number using function.
21. Write a python program to display the multiplication table of a given number.
22. Write a python program to find the factorial of a number using recursion
23. Write a python program to check whether a given string is palindrome or not.
24. Write a python program to count the number of times each character appears in a string.
25. Write a python program to compare two strings.

26. Write a python program which accepts a string, splits it and outputs the list of words in uppercase.
27. Write a python program which accepts a list of numbers and outputs their squares.
28. Write a python program to swap 2 numbers using tuples and function.
29. Write a python program to generate a list of random values using function.
30. Write a python program to find the median of a set of numbers using list.
31. Write a python program to convert hex to binary using dictionary.
32. Write a python program to generate a histogram of the letters in a string using dictionary.
33. Write a python program to copy contents of one file to another omitting any lines that begin with
34. Write a python program to open and write your roll no an list of marks to a file using pickling.
35. Micro project implementation.